

**RMO(INDIA) – 2010**

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
  - Ruler and compass allowed.
  - Answer all the questions.
1. Let ABCDEF be a convex hexagon in which the diagonals AD, BE, CF are concurrent at O. Suppose the area of triangle OAF is the geometric mean of those of OAD and OEF, and the area of triangle OBC is the geometric mean of those of OAB and OCD. Prove that the area of triangle OED is the geometric mean of those of OCD and OEF. [16]
  2. Let  $p_1(x) = ax^2 - bx - c$ ,  $p_2(x) = bx^2 - cx - a$  and  $p_3(x) = cx^2 - ax - b$  be three quadratic polynomials, where a, n and c are non zero real numbers. Suppose there exist a real number  $\alpha$  such that  $p_1(\alpha) = p_2(\alpha) = p_3(\alpha)$ , then prove that  $a = b = c$ . [17]
  3. Find the number of 4 digits number in base 10 having non zero digits and which are divisible by 4 but not by 8. [16]
  4. Find three distinct positive integers with the atleast possible sum such that the sum of the reciprocals of any two integers among them is an integral multiple of the reciprocal of third integers. [17]
  5. Let ABC be a triangle in which  $m\angle A = 60^\circ$ . Let BE and CF are bisectors of  $\angle B$  and  $\angle C$  with E on AC and F on AB. Let M be the reflection of A in the line EF. Prove that M lies on BC. [17]
  6. For each integer  $n \geq 1$ , define  $a_n = \left\lfloor \frac{n}{\sqrt{n}} \right\rfloor$ , where  $[n]$  denotes the largest integer not exceeding n, for any real number n. Find the number of all n in the set  $\{1, 2, 3, \dots, 2010\}$  for which  $a_n > a_{n+1}$ . [17]

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